Causality Assessment – Amoxicillin-Induced Itchy Rash

Case Summary

Parameter	Details
Patient	Raj
Suspected drug	Amoxicillin
Initial dose	500 mg TID
Indication	Throat infection
Drug started	01-Jun-2021
First reaction	Generalized itchy rash on 02-Jun-2021
Dose reduced	250 mg TID on 03-Jun-2021
Rash disappeared	04-Jun-2021
Dose increased	500 mg TID on 05-Jun-2021
Reaction recurred	Generalized itchy rash on 06-Jun-2021

Causality Assessment

Assessment criterion	Analysis	Result
1. Temporal relationship	Rash appeared 1 day after starting amoxicillin, which is a plausible temporal relationship.	Positive
2. Dechallenge	After reducing amoxicillin from 500 mg TID to 250 mg TID, the rash disappeared within 1 day.	Positive
3. Rechallenge	When the dose was increased again to 500 mg TID, the generalized itchy rash reappeared the next day.	Strongly positive
4. Alternative causes	No other drug or alternative cause is mentioned in the case.	No clear alternative cause
5. Dose-response relationship	Reaction occurred at 500 mg TID, disappeared after dose reduction, and recurred after increasing back to 500 mg	Positive

Final Causality Classification: CERTAIN / Very Strongly Related

The causality between amoxicillin and the generalized itchy rash is assessed as **CERTAIN**. The reaction occurred shortly after initiation of amoxicillin, disappeared following dose reduction, and recurred following re-escalation to the original dose. The positive rechallenge provides strong evidence that amoxicillin was responsible for the adverse drug reaction.

Suspected ADR: Amoxicillin-induced generalized pruritic rash

Causality: Certain

Type of evidence: Positive temporal relationship + dechallenge + positive rechallenge + dose-response relationship.